MATH329 · OPERATIONS RESEARCH

# Midterm 462 Practice

09/03/2025 · 30 marks

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## 01 Questions

Answer each question in the space below it. The full solution is in a separate file.

**Q1** THEORY · 5 M

Answer the following:

a) When solving a linear programming problem graphically, there are special cases that we encounter, mention them?

b) To start the simplex algorithm, there are two requirements that must be met. What are they?

**Q2** GRAPHICAL · 7 M

Solve the following problem graphically:

Maximum $z = 3x_1 + 2x_2$

S.to: $2x_1 + x_2 \leq 4$

$-2x_1 + x_2 \leq 2$

$x_2 \geq 1$

$x_1 \geq 0, x_2 \geq 0$

**Q3** STANDARD FORM · 8 M

Given the following problem:

Maximum $Z = 8x_1 - 2x_2$

S.to $2x_1 + x_2 \leq 12$

$x_1 - 5x_2 \geq 6$

$4x_2 = -7$

$x_1 \geq 0$, $x_2$ unrestricted in sign.

**1. What methods can be applied to solve the given linear programming problem?**

**2. Write the problem in standard form (equation form). "Don't solve the problem"**

**Q4** STANDARD FORM + SIMPLEX · 10 M

Given the following problem:

Min $Z = x_1 - 2x_2$

S.to $2x_1 + 3x_3 \leq 1$

$3x_1 + 2x_2 - x_3 \leq 5$

$x_1 \geq 0, x_2 \geq 0, x_3 \geq 0$

a) Write the problem in standard form (equation form).

b) Solve the LP problem using the Simplex method.

| Basic | $x_1$ | $x_2$ | $x_3$ | $s_1$ | $s_2$ | Sol. |
|---|---|---|---|---|---|---|
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